

T104 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

UNIT - VI

COMMUNICATION AND COMPUTER SYSTEMS

Model of communication system - Analog and digital - Wired and wireless channel. Block diagram of various communication systems - Microwave, satellite, optical fiber and cellular mobile system. Network model - PAN, LAN, MAN and WAN - Circuit and packet switching - Overview of ISDN.

Part-A (2 Marks)

1. What do you mean by WAN? (Jan 2014)

A wide-area network (WAN) is a collection of local-area networks (LANs) or other networks that communicate with one another. A WAN is essentially a network of networks, with the Internet the world's largest WAN.

A wide area network (WAN) is a network that covers a broad area (i.e., any telecommunications network that links across metropolitan, regional, national or international boundaries) using leased telecommunication lines.

2. What are the advantages of optical fiber communication? (April-2014)

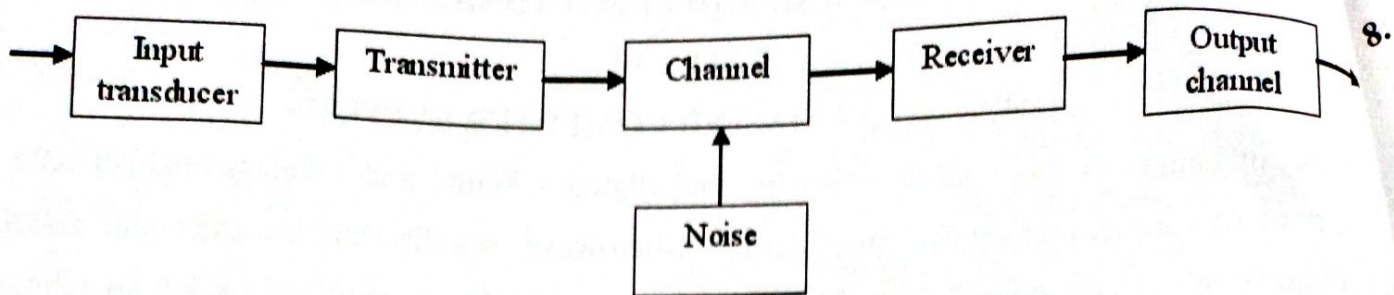
Advantages:

- More bandwidth
- Free from temperature variations
- Fiber dimension is very small
- Fiber is immune to chemical effects
- Highly secured hence used in useful in military communication
- Fiber is non-inductive , non-capacitive and non-conductive

3. What is LAN? (May 2015)

A local area network (LAN) supplies networking capability to a group of computers in close proximity to each other such as in an office building, a school,(or) a home. A LAN is useful for sharing resources like files, printers, games or other applications. A LAN in turn often connects to other LANs, and to the Internet (or) other WAN.

4. Sketch the block diagram of basic communication system. (Nov 2016, Dec-2018)



5. Define amplitude modulation. (May 2016, May-2018)

Amplitude modulation (AM) is a modulation technique used in electronic communication, most commonly for transmitting messages with a radio carrier wave. In amplitude modulation, the amplitude (signal strength) of the carrier wave is varied in proportion to that of the message signal, such as an audio signal.

6. Distinguish between AM and FM. (May 2016)

S.No.	FM	AM
1.	Amplitude of FM wave is constant. It is independent of the modulation index.	Amplitude of AM wave will change with the modulating voltage.
2.	Hence, transmitted power remains constant. It is independent of m_f .	Transmitted power is dependent on the modulation index.
3.	All the transmitted power is useful.	Carrier power and one sideband power are useless.
4.	FM receivers are immune to noise.	AM receivers are not immune to noise.
5.	It is possible to decrease noise further by increasing deviation.	This feature is absent in AM.
6.	Bandwidth = $2[\Delta f + f_m]$. The bandwidth depends on modulation index.	$BW = 2 f_m$. It is not dependent on the modulation index.
7.	BW is large. Hence, wide channel is required.	BW is much less than FM.

7. List the various types of network model. (April 2004, Nov 2016)

A computer network is mainly of four types:

- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)

8. What do you mean by MAN? (Nov 2017)

A metropolitan area network (MAN) is smaller than a wide area network (WAN) but larger than a local area network (LAN).

A metropolitan area network (MAN) is a computer network that usually spans a city or a large campus. A MAN usually interconnects a number of local area networks using a high-capacity backbone technology, such as fibre optical links, and provide up-link services to wide area networks and the Internet.

9. Brief about cellular or mobile services. (Nov-2018)

A terrestrial radio-based service providing two-way communications by dividing the serving area into a regular pattern of sub-areas or cells, each with a base station having a low-power transmitter and receiver.

10. What are the two levels of ISDN services? (Nov-2018)

There are two types of ISDN networks - BRI (Basic Rate Interface) and PRI (Primary Rate Interface). The major difference between BRI and PRI is the level of service and reliability.

11. Define baud rate. (May 2019)

The baud rate is the rate at which information is transferred in a communication channel.

12. Define satellite. (Nov 2019)

Satellite communication is the method of transporting information from one place to another using a communication satellite in orbit around the Earth. Telephone, radio, television, internet, and military applications use satellite communications.

13. What is microwave? (SEP 2020)

Electromagnetic waves on the frequency range of 1GHz to 30 GHz are referred to as microwaves.

A microwave link is a communications system that uses a beam of radio waves in the microwave frequency range to transmit video, audio, or data between two locations, which can be from just a few feet or meters to several miles or kilometers apart.

PART-B (10 Marks)

1. Draw the block diagram of microwave communication system and explain its operation. (Jan- 2014, Jan-2016, Dec-2018)

MICROWAVE COMMUNICATION

- Electromagnetic waves on the frequency range of 1GHz to 30 GHz are referred to as microwaves. Fig shows the block diagram of microwave communication.
- Several signal channels are modulated into micro-wave carrier frequencies and transmitted to the repeater stations spaced 50 to 70 km apart.
- As microwaves travel only on line-of-sight paths, the transmitter and receiver should be visible to each other. Hence, it is necessary to provide repeater stations in between the terminal stations at about 50km intervals.

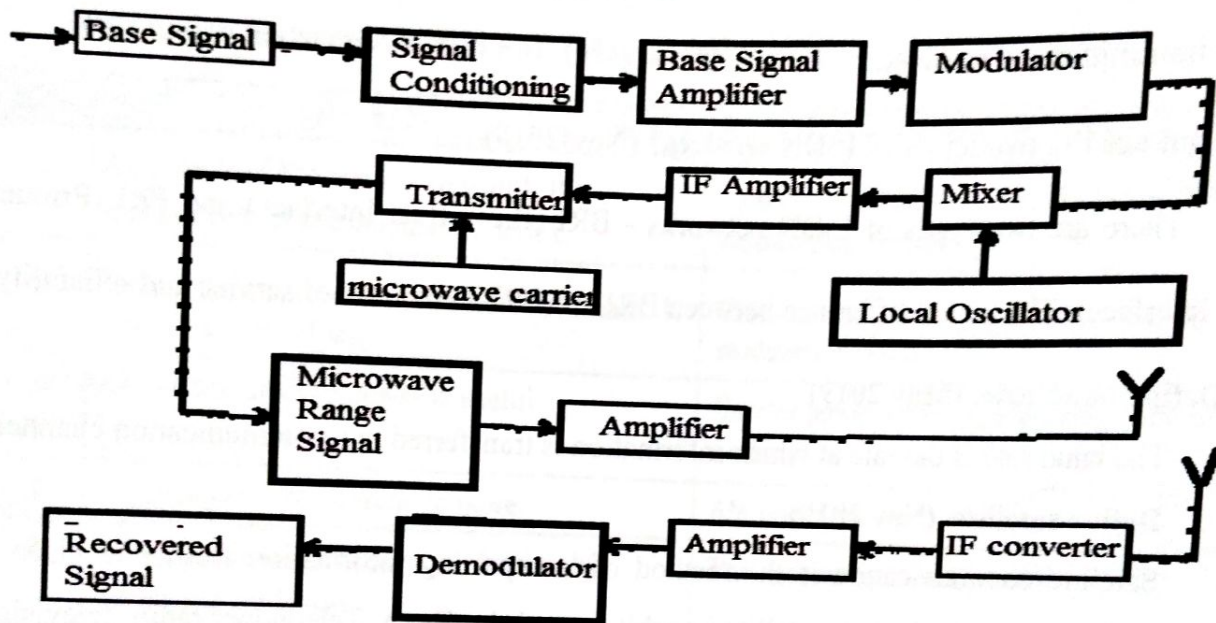


Fig Block Diagram of Microwave Communication

- Repeater antennas are placed on roads and hilltops so that there will not be any obstruction in the line of sight between the repeater antennas. This system is called Line of sight microwave link.
- As microwave communication offers a large transmission bandwidth, many thousands of telephone channels along with a few TV channels can be transmitted over the same route using the same facilities.

➤ Normally carrier frequencies in the 3 to 12GHz range are used for microwave communication.

➤ The transmitter output powers can be low because highly directional high gain antennas are used.

The general block diagram of line of sight transmission system is shown in Fig

The base signal is first conditioned using conditional circuits.

➤ The conditioned signal is amplified using amplifiers to increase the signal strength.

➤ The signal is fed into modulator to produce a frequency modulated wave.

➤ This signal is mixed with local oscillator signal.

➤ The output from the mixer is intermediate frequency(IF) wave and is fed into the transmitter where the IF wave is transferred into microwave range using microwave carrier.

➤ Then it is transmitted through the antennas after amplification.

➤ The receiver separates the IF signals and demodulates to recover the original data or message.

➤ Presently, microwave communications are widely used for telephone networks, in broadcast and television systems and in several other communications applications by services, railways, etc.

Properties of Microwave Links

- Involve line of sight(LOS) communication technology
- Affected greatly by environmental constraints, including rain fade
- Have very limited penetration capabilities through obstacles such as hills, buildings and trees

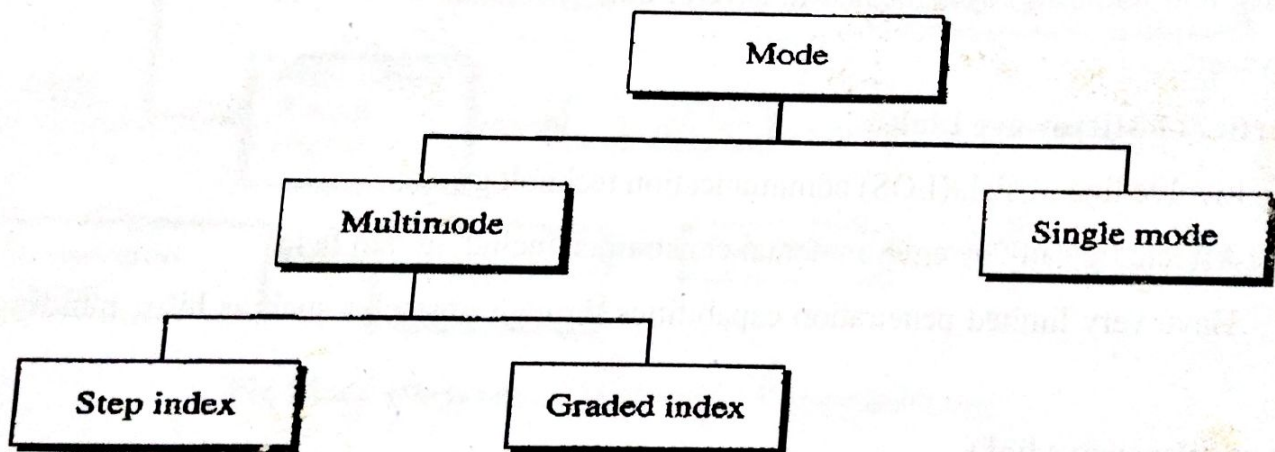
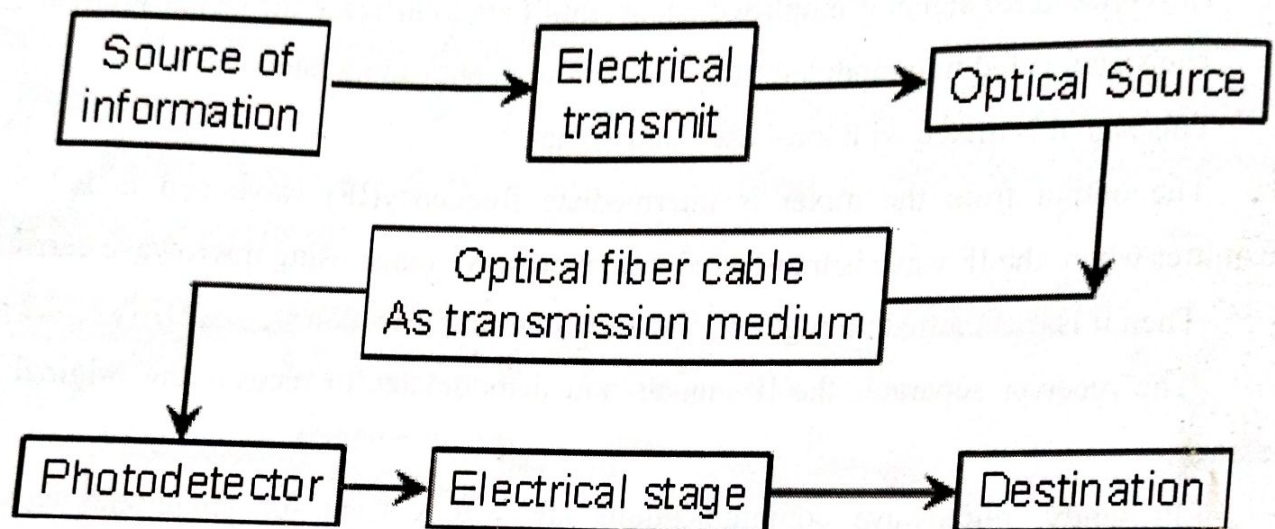
Uses of microwave links

- In communications between satellites and base stations
- As backbone carriers for cellular systems
- In short range indoor communications
- Telecommunications, in linking remote and regional telephone exchanges to larger (main) exchanges without the need for copper/optical fibre lines.
- Frequency ranges from 150 MHz to 150 GHz

2. With help of a neat diagram, explain the basic component of optical fiber communication system. (May 2015, Nov-2016, Nov-2017, May-2019, Nov-2019, Sept-2020)

Fiber-Optic Cable

- ✓ It is made of glass or plastic and transmits signals in the form of light
- ✓ Light travels in straight line through one substance
- ✓ If a ray of light travelling through one substance suddenly enters into another substance the ray changes its direction



- Electromagnetic waves in optical frequency range 10^{13} - 10^{15} Hz
- Fiber is a medium of transmission
- The light ray carrier carries the information and travels through fiber.
- An optical fiber consists of electrical transmitter, optical source, optical detector fiber and electrical receiver.
- The message to be transferred is first converted to electrical signal in electrical transmission block

- This electrical signal allowed to modulate the light from optical source
- The modulated light is launched into optical fiber which is the channel linking transmitter and receiver
- At the receiving end input optical signal is converted into electrical signal by the optical detector.

Advantages:

- More bandwidth
- Free from temperature variations
- Fiber dimension is very small
- Fiber is immune to chemical effects
- Highly secured hence used in useful in military communication
- Fiber is non-inductive , non-capacitive and non-conductive

Disadvantages:

- Termination of fiber optics is complex
- Cost is high
- Requires special tools

Applications:

- In medical electronics by using fiber noise free control signals are processed
- Computer communications
- Used in power plants
- Used in metal industry
- Used for long distance communication of speech, music videos and digital data signals
- Used in cable TV networks
- Wireless channel

3. Explain the block diagram of Integrated Services Digital Network.(May- 2015,Nov-2017, Nov-2019)

Integrated Service Digital Network (ISDN)

Integrated Services Digital Network "a network that provides end to end digital connectivity between users"

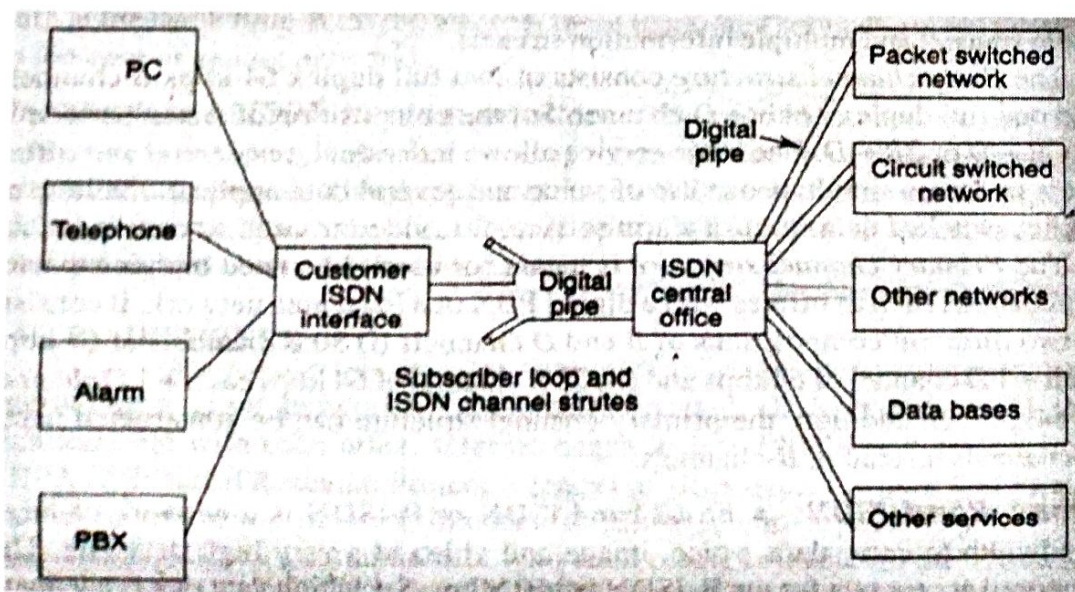
- ✓ ISDN provides services like direct dialing call wait, call hold etc...
- ✓ In direct dialing, the user can contact a person directly in an office by dialing extension number without the operator knowledge when called person is connected to private branch exchange (PBX)
- ✓ It consist of common physical interface, ISDN central office, digital subscriber loop, ISDN channels and integrated digital network
- ✓ ISDN is a new communication standard for providing an end to end connectivity over a Digital link. It is basically a circuit-Switched digital network. Conventional telephone lines that are used to carry the ISDN data are called ISDN lines in an ISDN line a wide range of digital Services such as voice, data, image, video, facsimile, etc can be integrated.

CCITT Consultative committee for international Telegraphy and Telephony introduced the first ISDN Standard in 1984. Called narrow-band ISDN. The Primary goal was to define an end-to-end digital interface.

Features of ISDN

- High Flexibility
- Local, National and international telephones.
- Video Conferencing
- Electronic data exchange.

COMMON PHYSICAL INTERFACE:



✓ In an ISDN, all devices such as telephone alarm, computer, PBX, LAN are connected to transmission line using a common physical interface.

✓ It essentially

provides a DTE-DCE connection where DTE- data terminal equipment, DCE-data communication equipment

Central office:

- ✓ It connects the numerous ISDN subscriber loop signals to the integrated digital network
- ✓ It provides subscribers to access to circuit switched network packet switched network databases and other networks

Digital subscriber loop and ISDN channels:

- It is the connection between physical interface and ISDN central office
- It has two different ISDN channel structures
- Basic channel structure
- Primary channel structure
- Both the channels are constructed from B-channel, D-channel, H -channel

B-channel:

- It is used to carry digital signals
- Circuit Switched and Packet switched can be constructed using this channel

D-channel:

- Used to carry signals to control circuit switched calls or to carry low speed data

H-channel:

- Used to carry high speed data
- E.g.: video, audio signals

Basic channel structure:

- It consists of B channel and D channel. The composition is 2B+D
- It allows individual, office, residential users to access several data applications

Primary channel structure:

- It is the combination of B and D channels
- There are 2 different types of composition i) 30Bchannel+1Dchannel

ii) 23 B channel+1D channel

It can also constructed using H-channel

Broad Band ISDN(B-ISDN)

It is a network of large bandwidth to carry data , voice , image , video at a very high data rate. Speed is 150Mbps

This high access rate allows broadcast traffic video conferencing etc.

ISDN System architectures

- The ISDN architecture consists of a customer's equipment and the interface between the customer and Telephone Company.
- The basic idea behind ISDN is that of the digital bit pipe. It is a conceptual pipe between the customer and the carrier through which bits flow. The bits can flow through this digital pipe in both the directions.
- The two principal standards used for bit pipe are: {i} a low bandwidth standard for home use, and {ii} a higher bandwidth standard for business use.

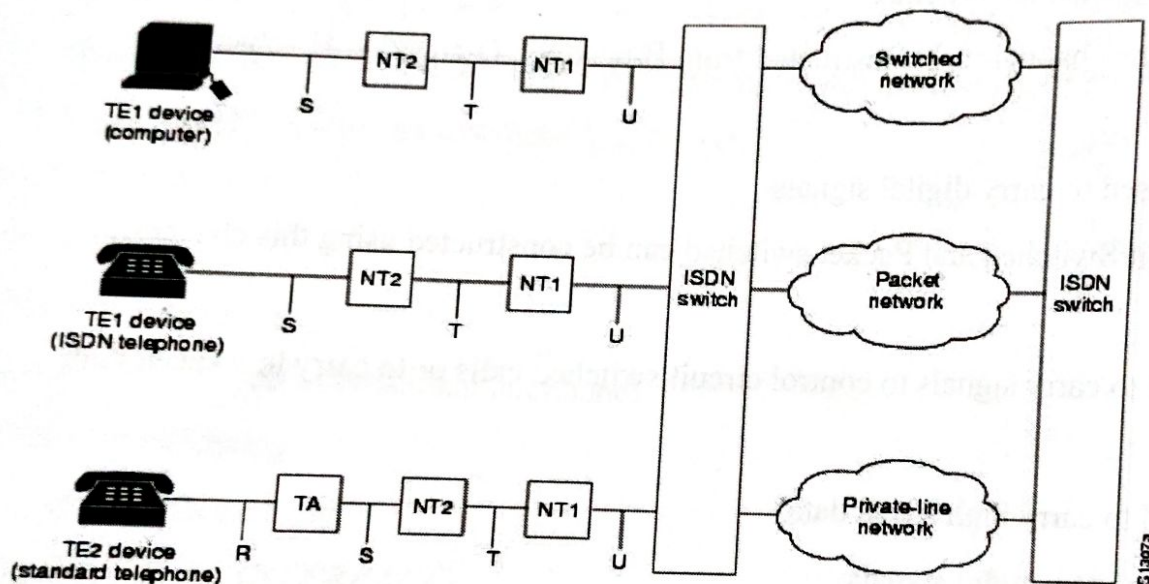


Fig Architecture of ISDN

- The carrier Places a Network Terminating device, NT1 on the customer's exchange in the carrier's office which is several kilometers away. Twisted pair is used for the connection.
- The NT1 box has a connector on it, into which a passive bus cable can be inserted. Up to 8 ISDN Telephones, terminals, alarms and other devices can be connected to the cable.
- In the above model, the device NT2 Called PBX (Private Branch exchange) is connected to NT1.

The four reference points in the above models R, S, T and U are defined below:

Basic Rate Interface

The entry level interface to ISDN is the Basic(s) Rate Interface (BRI), a 128kbit/s service delivered over a pair of standard telephone copper wires. The 128 kbit/s payload rate is broken down into two 64 kbit/s bearer channels ('B' channels) and one 16 kbit/s signaling channel ('D' channel or data channel). This is sometimes referred to as 2B+D.

The interface specifies the following network interfaces:

- The U interface is a two-wire interface between the exchange and a *network terminating unit*, which is usually the demarcation point in non-North American networks.
- The T interface is a serial interface between a computing device and a terminal adapter, which is the digital equivalent of a modem.
- The S interface is a four-wire bus that ISDN consumer devices plug into; the S & T reference points are commonly implemented as a single interface labeled 'S/T' on an Network termination 1 (NT1).
- The R interface defines the point between a non-ISDN device and a terminal adapter (TA) which provides translation to and from such a device.

BRI-ISDN is very popular in Europe but is much less common in North America. It is also common in Japan — where it is known as INS64.

Advantages of an ISDN line over a standard telephone line:

- Digital Service – Should have less errors
- Multiple Users
- Higher Bandwidth
- Connection is fast
- Permits use of newer techniques e.g. Video Phone.

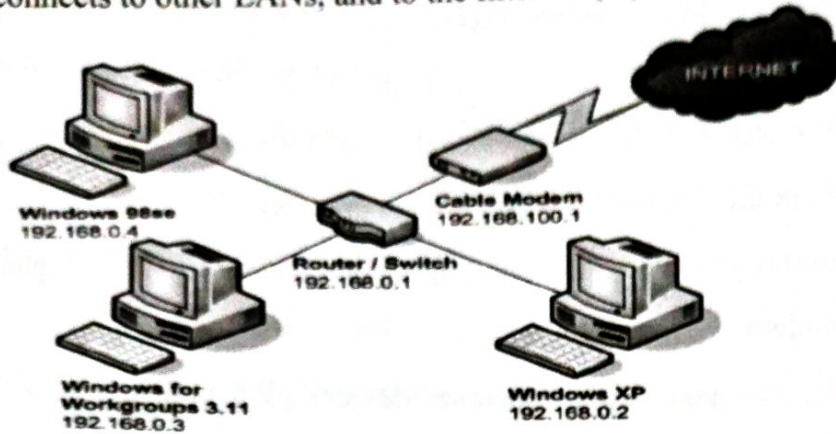
Applications using ISDN

- Wide area network
- Videotext
- Private network interconnection
- Telecommunication.

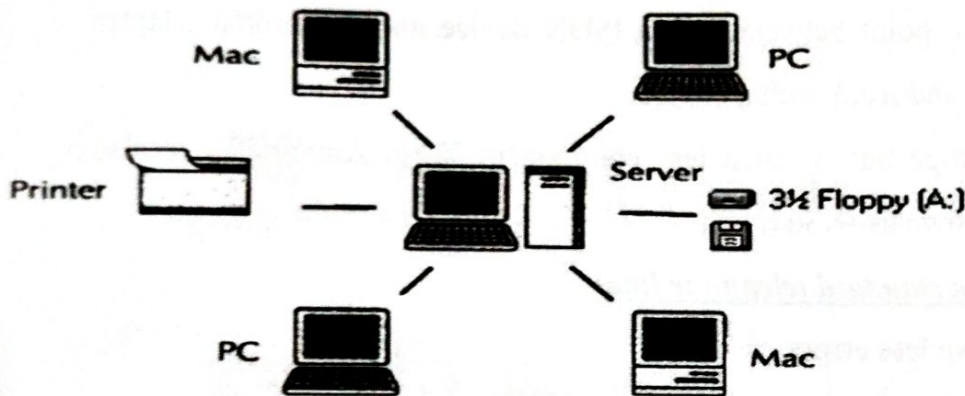
4. How the computer networks are classified? Explain them briefly. (JAN 2016)

Local area network (Jan-2014, May-2016, Dec-2018, May-2019)

A local area network (LAN) supplies networking capability to a group of computers in close proximity to each other such as in an office building, a school, (or) a home. A LAN is useful for sharing resources like files, printers, games or other applications. A LAN in turn often connects to other LANs, and to the Internet (or) other WAN.



Local Area Network (LAN)



Most local area network is built with relatively inexpensive hardware such as Ethernet cables, network adapters and hubs. Wireless LAN and other more advanced LAN hardware options also exist.

Fig: Local area Network.

- They are privately owned networks within a single building or campus up to a few km
- They are widely used to connect personal computers and workstation in company offices

and factories to share resources and exchange information

- LAN configuration consist of File server:
- Stores all the software's that control the network, as well as the software's that can be shared by the computers attached to the network

Workstation:

Computers connected to the file server

Cables:

Used o connect network interface cards in each computer

- Traditional LAN'S runs at a speed of 10Mbps to 100 Mbps
- Newer LAN'S operates at 10 Gbps

Advantages

- Easy to share devices
- Easy to share data
- Good bandwidth
- Data transfer is fast
- Low error rate
- Cost of setup is low

Disadvantages:

- Insufficient of power
- Maintenance is difficult
- Area covered is limited

WIDE AREA NETWORK (Dec-2018, Sept-2020)

A wide area network (WAN) is a network that covers a broad area (i.e., any telecommunications network that links across metropolitan, regional, national or international boundaries) using leased telecommunication lines.

Business and government entities utilize WANs to relay data among employees, clients, buyers, and suppliers from various geographical locations.

In essence, this mode of telecommunication allows a business to effectively carry out its daily function regardless of location.

The Internet can be considered a WAN as well, and is used by businesses, governments, organizations, and individuals for almost any purpose imaginable.

Related terms for other types of networks are personal area networks (PANs), local area networks (LANs), campus area networks (CANs), metropolitan area networks (MANs) which are usually limited to a room, building, campus or specific metropolitan area (e.g., a city) respectively.

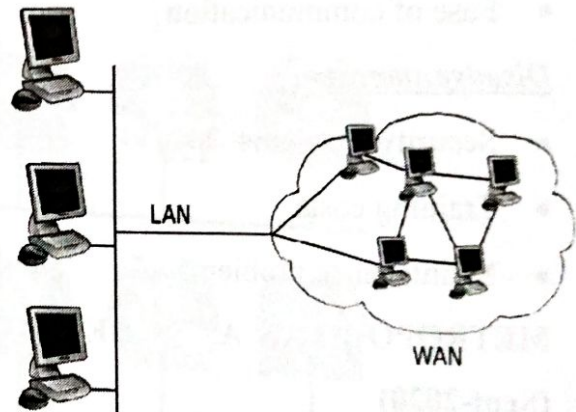
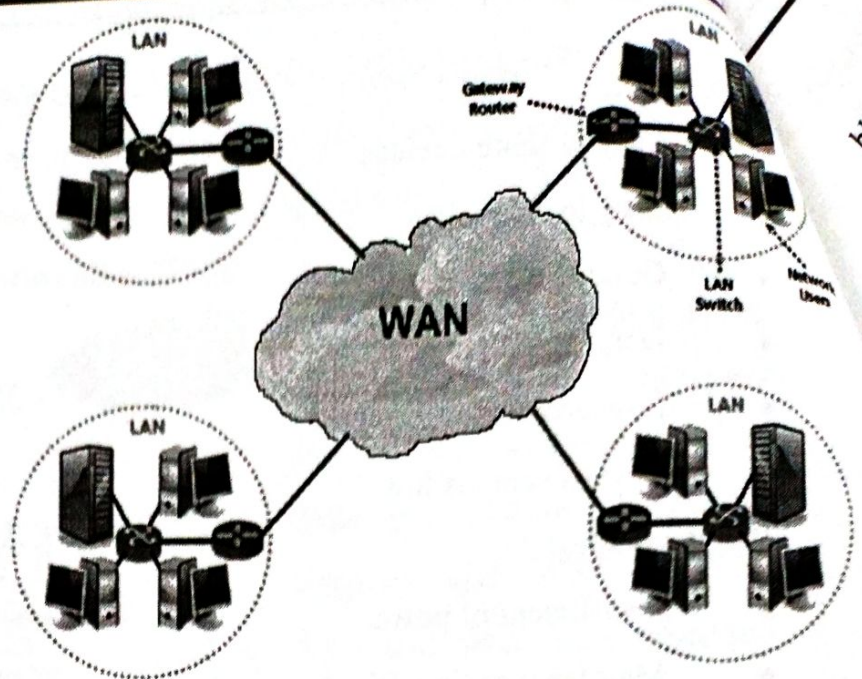


Fig: Wide area Network.

- Internet is a wide area network
- It is a collection of LAN'S
- It is used by big companies whether statewide or multinational
- They may run over leased lines
- They are collectively owned and maintained
- The fastest WAN is 155Mbps
- Leased phone lines uses 10Kbps and 2.5 Gbps



WAN - Wide Area Network

Advantages:

- High efficiency
- Low cost
- Ease of communication

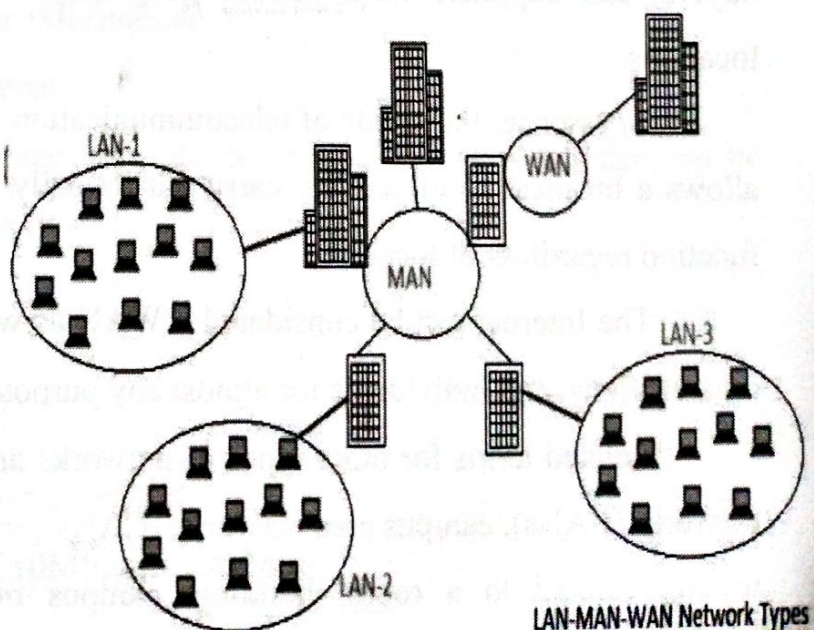
Disadvantages:

- Security problems
- Training costs
- Maintenance problems

METROPOLITAN AREA NETWORK

(Sept-2020)

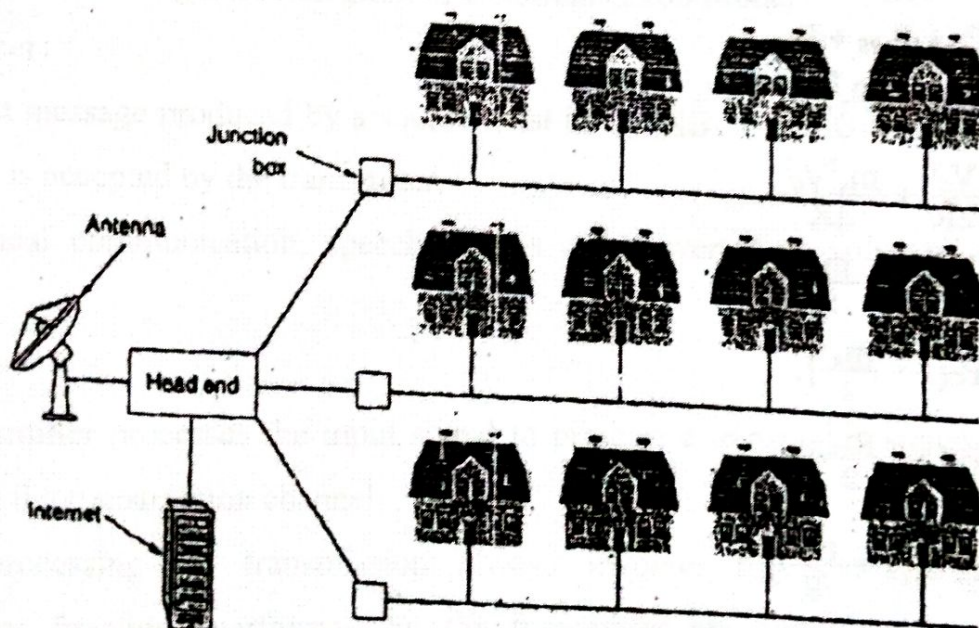
A metropolitan area network (MAN) is a computer network that usually spans a city or a large campus. A MAN usually interconnects a number of local area networks using a high-capacity backbone technology, such as fibre optical links, and provide up-link services to wide area networks and the Internet.



LAN-MAN-WAN Network Types

A MAN is optimized for a larger geographical area than a LAN, Ranging from several blocks of building to entire cities. MANs can also depend on communications channels of moderate-to-high data rates. A MAN might be owned and operated by a single organization, but it usually will be used by many individuals and organizations. MANs might also be owned and operated as public utilities. They will often provide means for internetworking of local networks.

- MAN covers large geographic areas
- It is often used in local libraries and government agencies to connect citizens and private industries
- A MAN is not generally owned by a separate organization
- Its communication link and equipments are generally owned by either of users or by a single network provider who sells service to the users.
- It operates at at a speed of 1Mbps.



Advantages:

- Good efficiency

Disadvantages:

- Cost is high

5. Describe the power relation of AM. (MAY 2016)

The total power in AM wave is equal to the sum of powers of the carrier, the upper sideband and the lower sideband.

$$P_T = P_C + P_{LSB} + P_{USB}$$

$$= \frac{V_C^2}{R} + \frac{V_{USB}^2}{R} + \frac{V_{LSB}^2}{R}$$

$$\text{Where } V_{LSB} = V_{USB} = \frac{m_a V_C}{2}$$

R is the resistance of antenna.

$$P_C = \frac{V_{rms}^2}{R} = \left(\frac{V_C}{\sqrt{2}} \right)^2 = \frac{V_C^2}{2R}$$

$$P_{USB} = P_{LSB} = \frac{V^2}{R} = \left(\frac{\frac{m_a V_C}{2}}{\sqrt{2}} \right)^2$$

$$= \frac{m_a^2 V_C^2}{8R}$$

$$\therefore P_T = P_C + P_{LSB} + P_{USB}$$

$$= \frac{V_C^2}{2R} + \frac{m_a^2 V_C^2}{8R} + \frac{m_a^2 V_C^2}{8R}$$

$$= \frac{V_C^2}{2R} + \frac{m_a^2 V_C^2}{4R}$$

$$P_T = \frac{V_C^2}{2R} \left[1 + \frac{m_a^2}{2} \right]$$

$$\therefore P_T = P_C \left[1 + \frac{m_a^2}{2} \right]$$

$$P_T = P_C \left[1 + \frac{m_a^2}{2} \right]$$

$$I_r^2 R = I_c^2 R \left[1 + \frac{m_a^2}{2} \right]$$

$$I_r^2 = I_c^2 \left[1 + \frac{m_a^2}{2} \right]$$

$$I_r = \sqrt{I_c^2 \left[1 + \frac{m_a^2}{2} \right]}$$

$$I_r = I_c \sqrt{1 + \frac{m_a^2}{2}}$$

1. Define communication system. Explain with neat block diagram of wireless communication system. (10). (NOV 2016)

Communication is the transfer of information from one place to another place.

Distribution of data, message (or) information from one location to another location with high reliability and security is the role of communication system.

Elements of Communication System (Communication Model):

Communication system consists of the following components which acts together to accomplish information transfer or exchange.

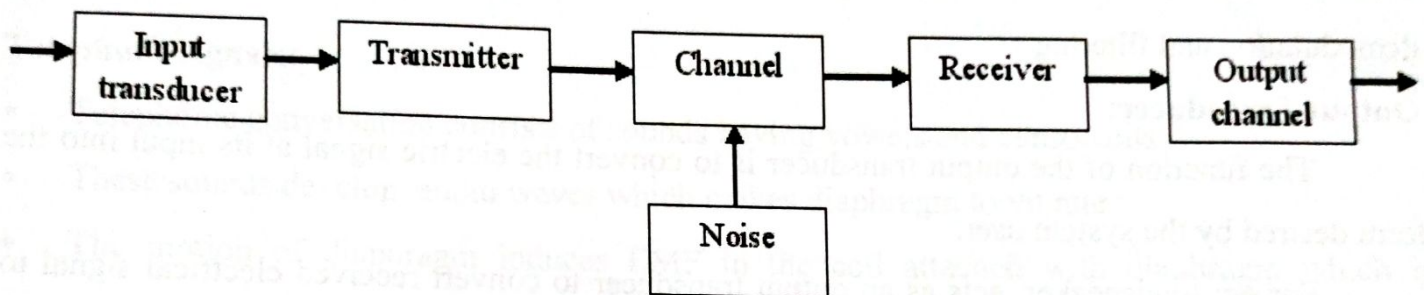


Fig Block diagram of Communication Model

Input Transducer:

The input message produced by a source must be converted by a transducer to a suitable form such that it is accepted by the transmitter.

In electrical communication, speech signals are converted to voltage variation by a microphone.

Transmitter:

The transmitter processes the input signal to produce a transmitted signal suited to the characteristics of the transmission channel.

Signal processing for transmission always involves modulation. In addition to modulation, other functions performed by the transmitter are amplification, filtering and coupling the modulated signal to the channel.

Channel:

The channel can have different forms: the atmosphere or free space, coaxial cable, fiber optic, waveguide, etc.

The signal undergoes some amount of degradation in the channel due to interference and distortion resulting from band limitation and nonlinearities.

Types of channel:

Wired – eg: Twisted pair, coaxial cable, optical fiber

Wireless – eg: microwave, satellite, mobile

Receiver:

- The receiver function is to extract the desired signal from the received signal at the channel output and to convert it to a form suitable for the output transducer.
- Other functions of the receiver are amplification (the received signal may be extremely weak), demodulation and filtering.

Output Transducer:

The function of the output transducer is to convert the electric signal at its input into the form desired by the system user.

For eg: loudspeaker, acts as an output transducer to convert received electrical signal to voice signal.

7. Describe different types of signals and principles of AM and FM. (MAY 2018)

Types of Signals

Signals are defined as “physical phenomenon (quantity) that varies with respect to time

Data signals may be classified as

- ✓ Analog signals
- ✓ Digital signals

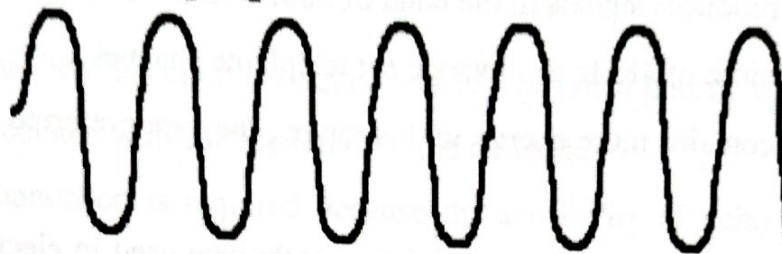
Analog and Digital Communication System

- Analog signals are the one which varies continuously with respect to time
- There are many kinds of information sources, which can be divided into distinct message categories, analog and digital.
- An analog message is a physical quantity that varies with time, usually in a smooth and continuous fashion.
- Since the information resides in a time-varying waveform, an analog communication system should deliver this waveform with a specified degree of reliability or fidelity.
- They are sinusoidal in nature

- They are represented by voltage waveforms that have different amplitudes at different instant of time

e.g.: telephone, radio broadcast and TV signals.

Analog signal



Telephone signals:

- Telephonic conversation consists of sounds having vowels and consonants
- These sounds develop audio waves which makes diaphragm to vibrate
- The motion of diaphragm induces EMF in the coil attached with diaphragm which is surrounded by static magnetic field on all sides
- Induced EMF is electrical equivalent of sound waves and this may be broken into sinusoidal signals at different frequencies
- For every telephone channel the overall bandwidth needed is 4KHz

Digital signals:

- It is a discrete time signal
- It plays a significant role in transmission and receiving the coded signals
- A digital message is an ordered sequence of symbols selected from a finite set of discrete elements.
- Since the information resides in discrete symbols, a digital communication system should deliver these symbols with a specified degree of accuracy in a specified amount of time.
- E.g: telegraph ,radar signals , teleprompter signals

Digital signal



TELEPHONY:

- It is a communication service for the transmission of speech signal between two points
- Speech signal is converted to electrical signal by microphone
- The electrical signal then transmitted through telephone line to a distant receiver
- Human voice produces signals in the band of 300Hz to 3.4KHz
- Hence a bandwidth of 4KHz is allocated for telephone channel
- Female voice contains more energy in this range, they are preferred for telephone operators and announcers

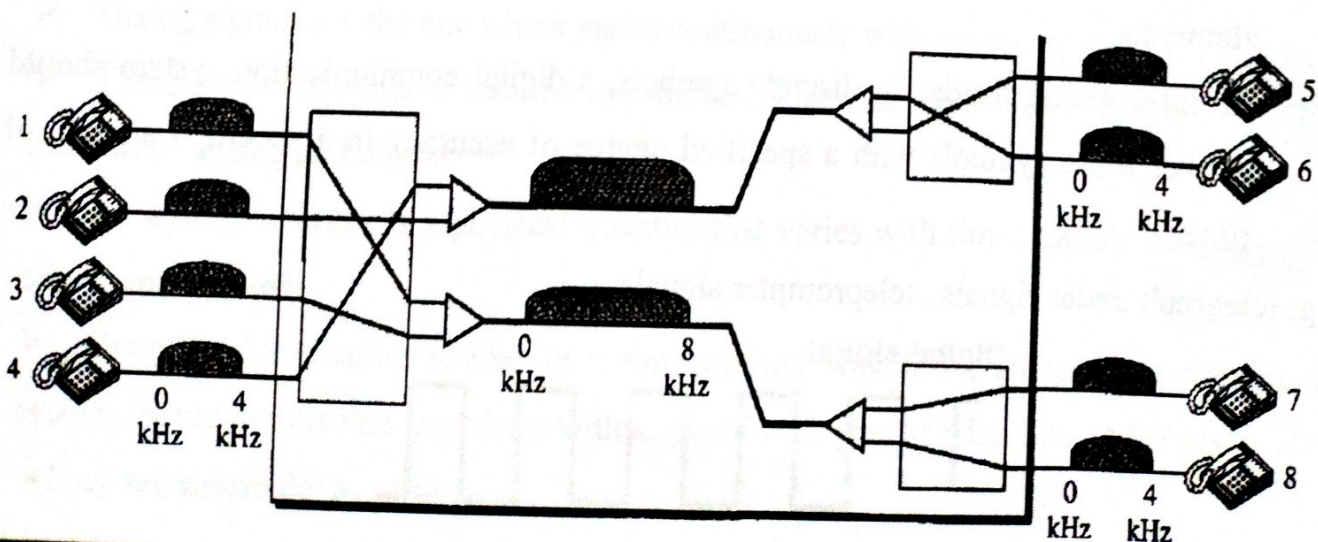
Amplitude modulation (AM) is a modulation technique used in electronic **communication**, most commonly for transmitting messages with a radio **carrier** wave. In amplitude modulation, the amplitude (signal strength) of the **carrier** wave is varied in proportion to that of the message signal, such as an audio signal.

Frequency modulation (FM) is the encoding of information in a carrier wave by varying the instantaneous frequency of the wave. The technology is used in telecommunications, radio broadcasting, and signal processing, and computing.

8. Write a short note on circuit and packet switching. (May 2018)

Circuit Switching:

- Circuit Mode is the most familiar type of switching to most people.
- In a circuit switched network, the communication pathway between two users is fixed for the duration of the call and is not shared by other users.
- Although several users may share one physical line by using equipment that support frequency division multiplexing.

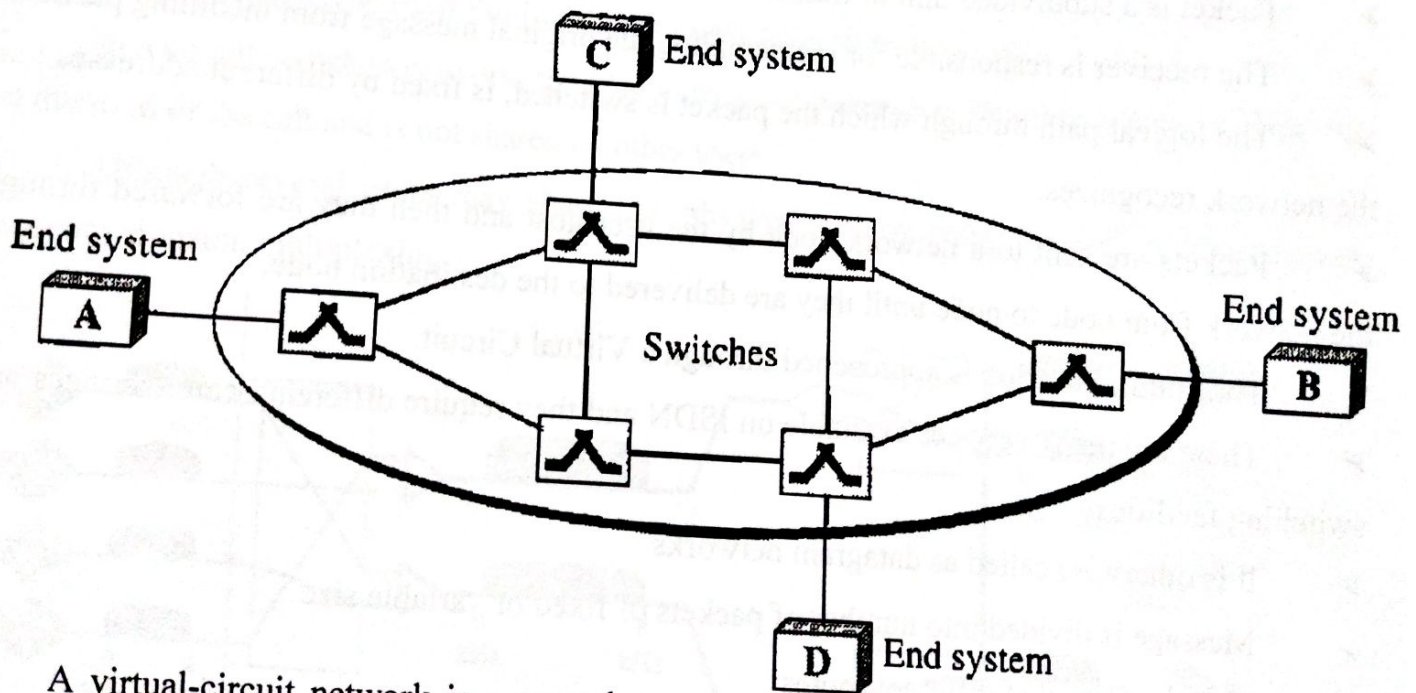
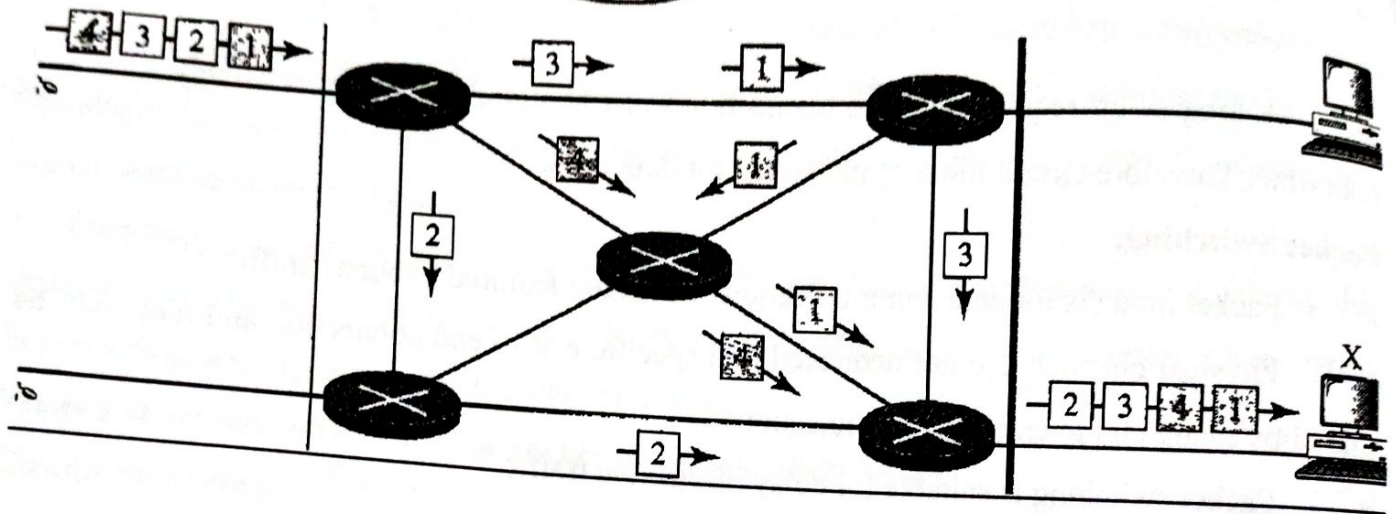
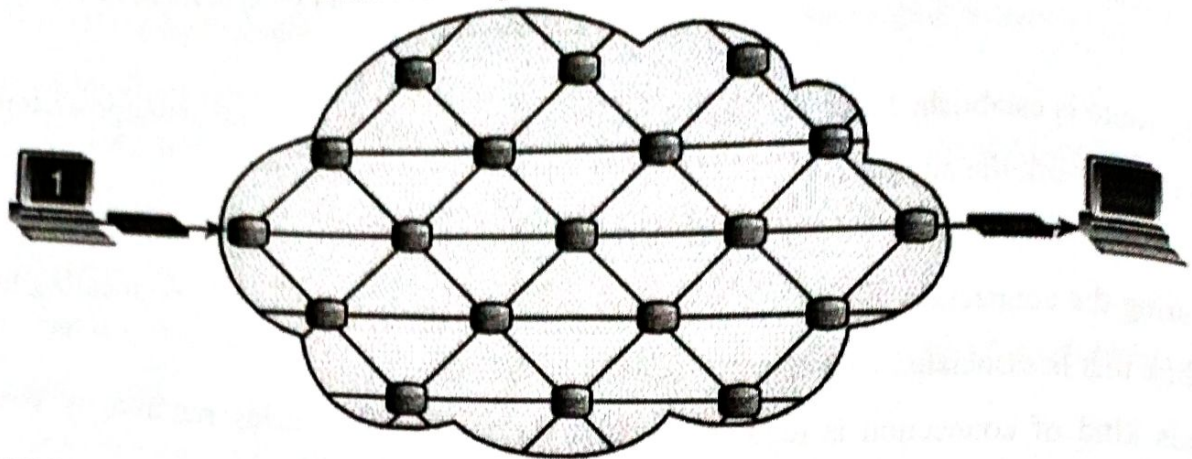


- In circuit mode, a communication is obtained between two users by establishing a fixed pathway.
- The route is established after the calling initiates the call initiates the call setup procedure by giving the network the addresses.
- During the connection, the circuit is equivalent to a physical pair of wires connecting the two user link that is associated with connection oriented network.
- This kind of connection is required because the sensitivity of delay natured by voice calls.
- A long setup required by this connection must be compensated by relatively long call hold time. Therefore circuit mode is not suited for data calls.

Packet Switching:

- Packet mode is for data communications which are featured by bursty traffic.
- Physical channels are not dedicated to a specific end to end connection and they may be shared by many end to end logical connections.
- Packet switching is suitable for delay insensitive traffic.
- Packet is a subdivided unit of transmission data.
- The receiver is responsible for reassembling the original message from incoming packets.
- The logical path through which the packet is switched, is fixed by different addresses that the network recognizes.
- Packets are sent to a network node by the user host and then they are forwarded through the network from node to node until they are delivered to the destination node.
- The communication is approached through a Virtual Circuit.
- There are many services available on ISDN and they require different bearer services and switching facilities.
- It is otherwise called as datagram networks
- Message is divided into number of packets of fixed or variable size
- They can travel at different routes
- The routes are selected based on the switches

Packet routing through WAN/Internet



- A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

- As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
- Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
- As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction (it defines what should be the next switch and the channel on which the packet is being carried), not end-to-end jurisdiction. The reader may ask how the intermediate switches know where to send the packet if there is no final destination address carried by a packet.
- As in a circuit-switched network, all packets follow the same path established during the connection.
- A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer.

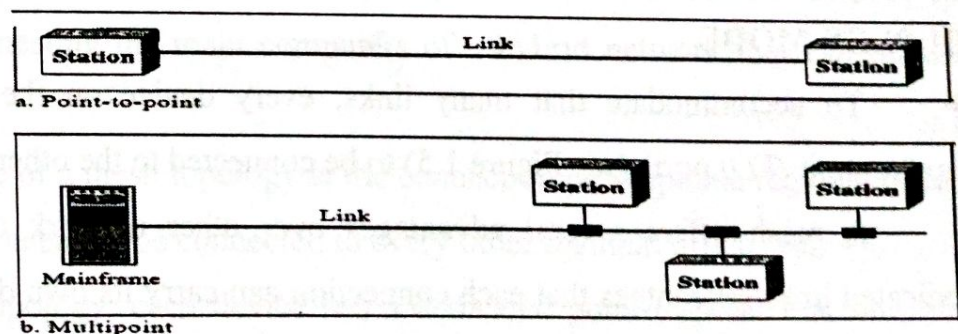
CIRCUIT SWITCHING	PACKET SWITCHING
•Dedicated transmission path	•No dedicated transmission path
•Messages are not stored	•Packets must be stored
•Path established for entire conversation	•Route established by each packet
•Fixed bandwidth	•Dynamic bandwidth

9. Write short notes on the following (MAY 2019)

a. Topology (4)

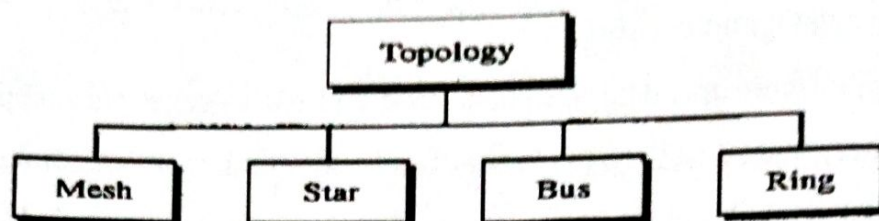
Physical Topology

- The term *physical topology* refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology.



Types of connections: point-to-point and multipoint

- The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.



Categories of Topology

- There are four basic topologies possible: mesh, star, bus, and ring of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.
- There are four basic topologies possible: mesh, star, bus, and ring (see Figure 1.4).

Mesh

- In a mesh topology, every device has a dedicated point-to-point link to every other device. The term *dedicated* means that the link carries traffic only between the two devices it connects.
- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, $n(n - 1) / 2$.

DUPLEX-MODE

- To accommodate that many links, every device on the network must have $n - 1$ input/output (VO) ports (see Figure 1.5) to be connected to the other $n - 1$ stations.
- A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

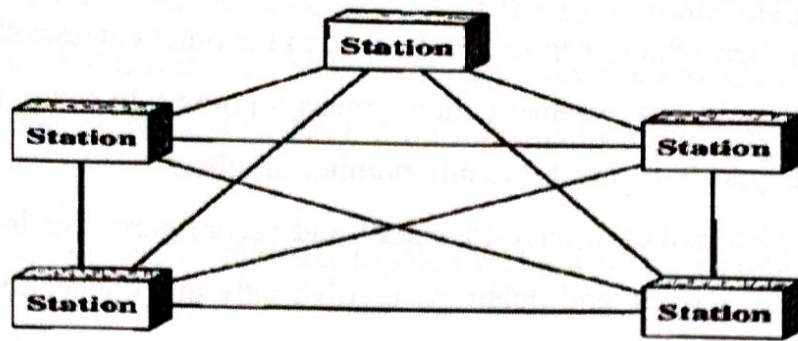


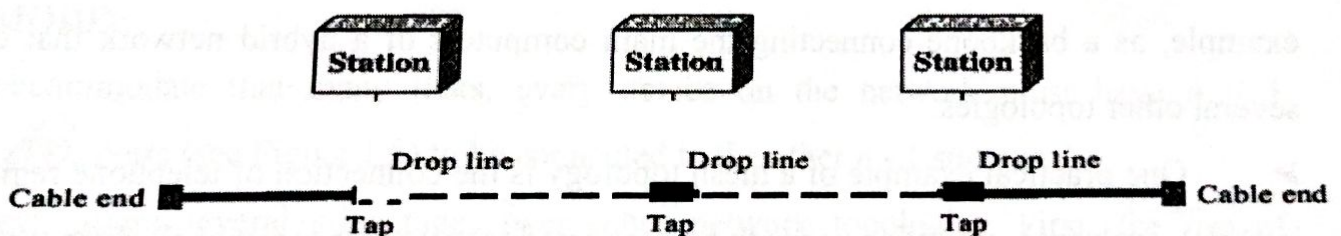
Figure A fully connected mesh topology (five devices)

- Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- Finally, point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.
- The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult.
- Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.
- For these reasons a mesh topology is usually implemented in a limited fashion, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.
- One practical example of a mesh topology is the connection of telephone regional offices in which each regional office needs to be connected to every other regional office.
- **Star Topology** In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.

- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.
- A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others.
- This factor also makes it easy to install and reconfigure. Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.
- Other advantages include robustness. If one link fails, only that link is affected. All other links remain active. This factor also lends fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.
- One big disadvantage of a star topology is the dependency of the whole to polygon on one single point, the hub. If the hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies.
- The star topology is used in local-area networks (LANs). High-speed LANs often use a star topology with a central hub.

BUS TOPOLOGY

- The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint.
- Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. Advantages of a bus topology include ease of installation.



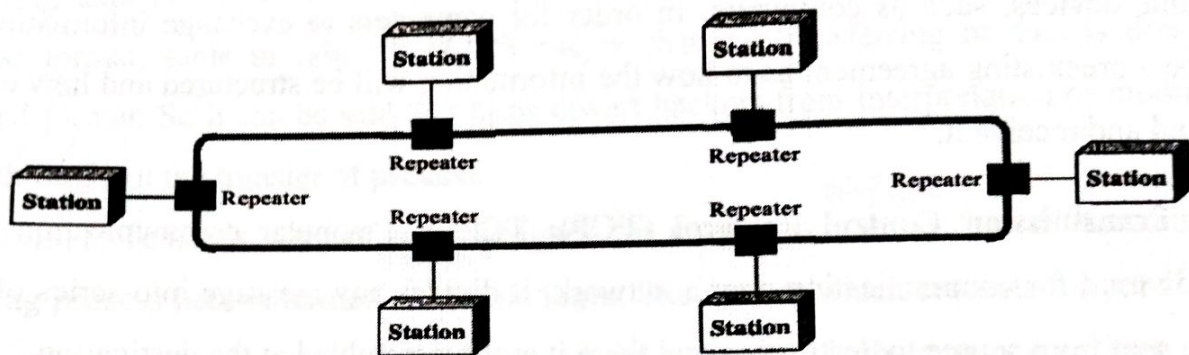
A bus topology connecting three stations

- Backbone cable can be laid along the most efficient path, and then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

- In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated. Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.
- Bus topology was the one of the first topologies used in the design of early local area networks.

RING TOPOLOGY

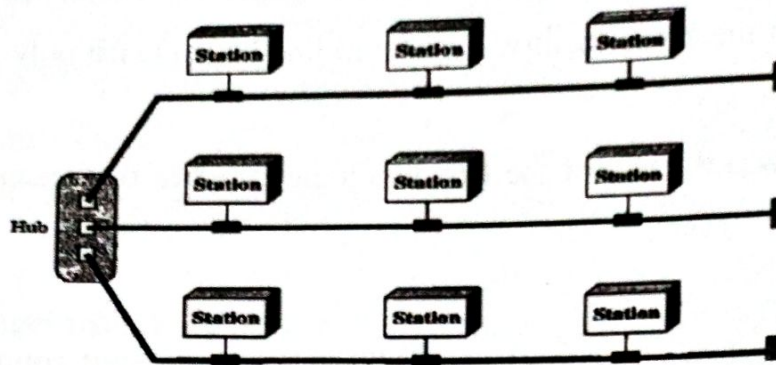
- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along



A ring topology connecting six stations

- A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections.
- The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified. Generally in a ring, a signal is circulating at all times.
- If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.
- However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network.

- This weakness can be solved by using a dual ring or a switch capable of closing off the break.



A hybrid topology: a star backbone with three bus networks

NETWORK MODELS

b. Protocol (3)

Protocol, in computer science, a set of rules or procedures for transmitting data between electronic devices, such as computers. In order for computers to exchange information, there must be a preexisting agreement as to how the information will be structured and how each side will send and receive it.

- **Transmission Control Protocol (TCP):** TCP is a popular communication protocol which is used for communicating over a network. It divides any message into series of packets that are sent from source to destination and there it gets reassembled at the destination.
- **Internet Protocol (IP):** IP is designed explicitly as addressing protocol. It is mostly used with TCP. The IP addresses in packets help in routing them through different nodes in a network until it reaches the destination system. TCP/IP is the most popular protocol connecting the networks.
- **User Datagram Protocol (UDP):** UDP is a substitute communication protocol to Transmission Control Protocol implemented primarily for creating loss-tolerating and low-latency linking between different applications.
- **Post office Protocol (POP):** POP3 is designed for receiving incoming E-mails.
- **Simple mail transport Protocol (SMTP):** SMTP is designed to send and distribute outgoing E-Mail.

- **File Transfer Protocol (FTP):** FTP allows users to transfer files from one machine to another. Types of files may include program files, multimedia files, text files, and documents, etc.
- **Hyper Text Transfer Protocol (HTTP):** HTTP is designed for transferring a hypertext among two or more systems. HTML tags are used for creating links. These links may be in any form like text or images. HTTP is designed on Client-server principles which allow a client system for establishing a connection with the server machine for making a request. The server acknowledges the request initiated by the client and responds accordingly.
- **Hyper Text Transfer Protocol Secure (HTTPS):** HTTPS is abbreviated as Hyper Text Transfer Protocol Secure is a standard protocol to secure the communication among two computers one using the browser and other fetching data from web server. HTTP is used for transferring data between the client browser (request) and the web server (response) in the hypertext format, same in case of HTTPS except that the transferring of data is done in an encrypted format. So it can be said that https thwart hackers from interpretation or modification of data throughout the transfer of packets.
- **Telnet:** Telnet is a set of rules designed for connecting one system with another. The connecting process here is termed as remote login. The system which requests for connection is the local computer, and the system which accepts the connection is the remote computer.
- **Gopher:** Gopher is a collection of rules implemented for searching, retrieving as well as displaying documents from isolated sites. Gopher also works on the client/server principle.

10. What is earth station? Explain the satellite communication system with the relevant block diagram. (May-2017)

Satellite Communication:

- The satellite is essentially a microwave station placed in orbital space.
- The broadcast signals are beamed up to the satellite from an earth station through the directional dish antenna that is synchronized to the position of the satellite; devices called transponder is used in the satellite to receive the weak signals and are amplified.
- The amplified signal further retransmits the back to another earth station in a different location on the earth.

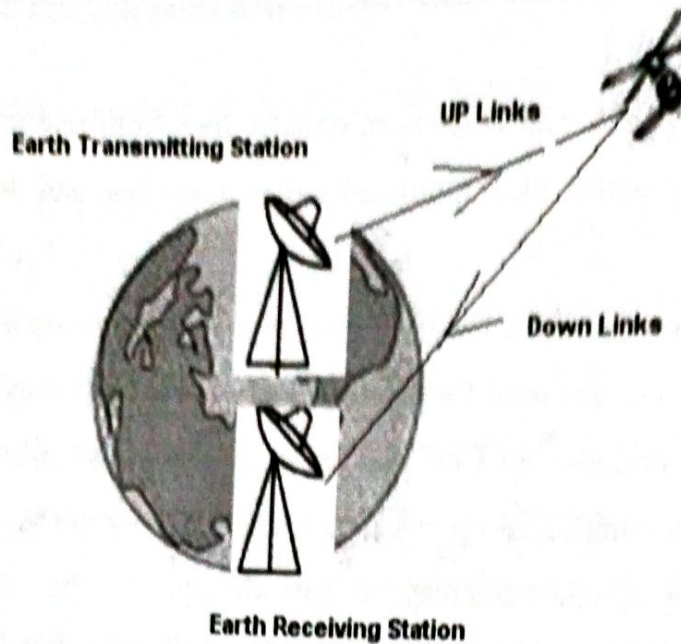


Fig. General Diagram of Satellite Systems

➤ The earth stations typically transmit their signals to satellites on carrier frequencies ranging from 5.92 to 6.43 GHz band called the **uplink frequency**.

➤ The satellite transponders convert these

signals to a range from 3.7 to 4.2 GHz band. These frequencies are referred to as **downlink frequencies**.

- In this orbit, the satellite is made to travel at a velocity equal to the rotation of earth in order to maintain a fixed position relative to a point about equator. A passive satellite simply reflects a signal back to earth and there are no gain devices on board to amplify the signal.
- On the other hand, an active satellite receives, amplifies and retransmits the signal back towards earth.

Orbital Patterns:

Once launched, a satellite remains in orbit because the centrifugal force caused by its rotation around the earth is counter balanced by the earth's gravitational pull.

The closer to earth the satellite rotates, the greater the gravitational pull, and greater the velocity required to keep it from being pulled to earth.

Low Altitude Satellites:

Close to earth (150 to 500km in height) travel at approx 28,500km/Hr.

At this speed, it takes approx 90minutes to rotate around the entire earth.

Consequently, the time that the satellite is in the Line of sight (LOS) of a particular earth station is 15 minutes or less per orbit.

Medium Altitude Satellites

9,500 to 19,000 km in height have a rotation period of 5 to 12 Hrs and remain in LOS of a particular earth station for 2 to 4 Hrs per orbit.

High Altitude Satellites

High altitude, geosynchronous satellite is a satellite which is placed at a height of 35,786 km from the earth's surface and has an orbital velocity equal to that of the earth.

A geosynchronous satellite that lies on the earth's equatorial plane is called as the geostationary satellite.

A geo-stationary satellite remains in a fixed position with respect to a given earth station and has 24hrs availability time.

Three geostationary satellites spaced 120° apart can cover the whole world.

Based on the area of the coverage, orbits are classified as

- ✓ **Polar orbit:** When the satellite rotates in an orbit that takes it over the north and south poles, it is called polar orbit.
- ✓ **Equatorial Orbit:** When the satellite rotates in an orbit above the equator, it is called an equatorial orbit.
- ✓ **Inclined Orbit:** Any other orbital path is called an inclined orbit.

Satellite System

A satellite system, consists of three basic sections,

The uplink (transmitting earth station)

The satellite transponder

The downlink (receiving earth station)

Typical frequencies for telecommunication services in a satellite system are 6/4 GHz and 14/12 GHz, where 6 and 14 GHz represent uplink frequencies and 4 and 12 GHz represent downlink frequencies.

Uplink Model:

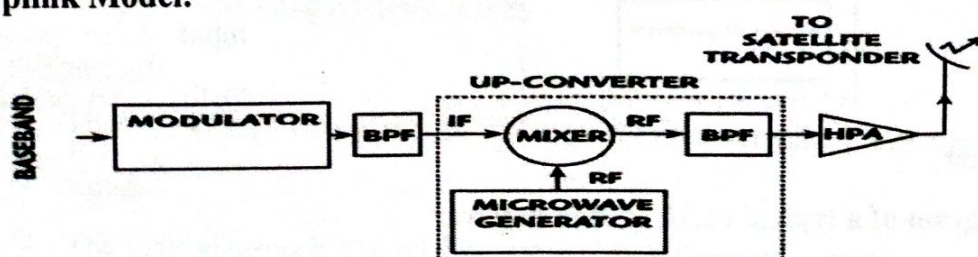


Fig. Block diagram of Earth Station Transmitter

A typical earth station transmitter consists of the following as shown in fig.

IF modulator

It converts the input base band signals to either an FM, or a PSK modulated intermediate frequency.

IF to RF microwave up converter

It translates the IF to appropriate RF carrier frequency.

High power amplifier: It provides adequate output power to propagate the signal to the satellite transponder.

Satellite Transponder: (The transponder is an RF-to RF repeater)

A Satellite Transponder consists of

- ✓ a Band Pass Filter(BPF)
- ✓ an input low noise amplifier(LNA)
- ✓ a frequency translator
- ✓ and a high power amplifier

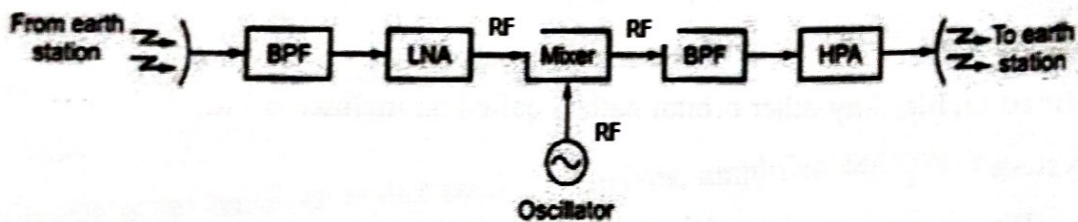
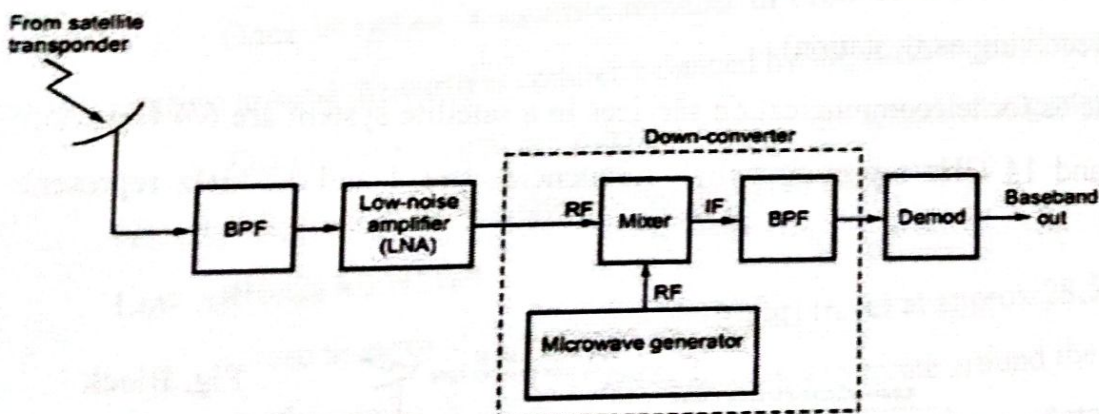


Fig. Block diagram of the satellite transponder

Downlink Model



RF-to-IF down converter

Fig. Block diagram of a typical earth station receiver

It is a mixer BPF-combination which converts the received RF signal to an IF frequency.

Communication Satellites

Communications Satellites are usually composed of the following subsystems:

- Communication Payload, normally composed of transponders, antenna, and switching systems
- Engines used to bring the satellite to its desired orbit
- Station Keeping Tracking and stabilization subsystem used to keep the satellite in the right orbit, with its antennas pointed in the right direction, and its power system pointed towards the sun
- Power subsystem, used to power the Satellite systems, normally composed of solar cells, and batteries that maintain power during solar eclipse
- Command and Control subsystem, which maintains communications with ground control stations. The ground control earth stations monitor the satellite performance and control its functionality during various phases of its life-cycle.

The bandwidth available from a satellite depends upon the number of transponders provided by the satellite. Each service (TV, Voice, Internet, radio) requires a different amount of bandwidth for transmission. This is typically known as link budgeting and a network simulator can be used to arrive at the exact value.

INTELSAT (USA based International Telecommunication Satellite Organization which provide services to 119 member countries)

INSAT (Indian Satellites which provide services to Indian region)

Advantages

- It provides point to multipoint communication
- Offers telecommunication links which includes telephone, TV, telegraphy, telex, FAX, video conferencing, video text, digital transmission services, etc.
- Global Positioning Systems (GPS)

Disadvantages

- There is a large time-delay of 250 millisecs between the transmission and reception of a signal.
- The malfunctions in the satellite are highly difficult to correct.
- The initial cost involved is quite large.